

Newton: 10× Saturn V Rocket Concept – Executive Summary

A **Newton rocket** is conceived as a hypothetical launch vehicle scaled 10× in every linear dimension from the Saturn V. Saturn V itself was ~111 m tall, 10 m in diameter, three stages, and a mass of $\sim 2.8\text{--}3.0 \times 10^6$ kg [5†L225-L228]. At 10× linear scale, Newton would be ~1 110 m tall and ~100 m in diameter, with ~1 000× the volume and roughly 1 000× the mass ($\sim 3 \times 10^9$ kg). Thrust must similarly scale up by ~1 000× to achieve the same launch thrust-to-weight, implying $\sim 3.4 \times 10^4$ MN of lift-off thrust (versus Saturn V's 34.5 MN [5†L260-L268]). These extreme values drive all other design considerations.

Newton's **structural** design must handle 1 000× greater gravity loads and bending moments than Saturn V. Buckling and aeroelastic instability become critical: scaling arguments show that, if wall thickness also scales 10×, the buckling capacity grows by ~100× (because moment of inertia $\propto L^4$ and length $\propto L$), yet weight grows 1 000×, so **buckling margin falls by ~10×**. This requires heavy reinforcement (thicker walls or grid-stiffening) far beyond historic practice. Engineers would likely use ring stiffeners and advanced materials (high-strength aluminium–lithium alloys or composites in non-critical areas) to raise buckling loads. Aerodynamic stability also demands careful analysis; Newton's slenderness ratio (~11:1 height:diameter) is similar to Saturn V's, but the absolute length (1 110 m) implies very low bending-mode frequencies. Active controls (engine gimbals, thrust vectoring) and possibly *in-flight* damping systems would be needed to suppress bending and flutter.

Propulsion must be massively larger. Saturn V's first stage used five 6.7 MN F-1 kerosene engines (LOX/RP-1) for 33 MN sea-level thrust [5†L260-L268]. No single-chamber engine has exceeded ~9 MN thrust (NASA's F-1A) [27†L132-L141]. Modern high-thrust engines (e.g. SpaceX Raptor) deliver only ~2–3 MN each [54†L364-L370]. Consequently, Newton must cluster **thousands** of engines. For example, using 3 MN Raptor-class engines would require ~11 000 engines to reach ~34 000 MN thrust – even optimistic future Raptor3 (3 MN) yields thousands of engines. Russian RD-170/171 uses four combustion chambers (7.9 MN total) instead of one for similar thrust [27†L144-L152], but Newton-level thrust likely needs a heterogeneous approach. Propellant choice also affects design: kerosene/LOX (Saturn V first stage) is dense and storable but low-Isp (~300 s), whereas LH₂/LOX gives high Isp (~450 s vacuum [5†L278-L282]) but huge tanks (Saturn's hydrolox stages were already 24–25 m long). A mixed approach (kerolox first stage, hydrolox upper stages) might be retained, or LOX/methane (CH₄) engines (Raptor/BE-4) could be adopted as a compromise.

Manufacturing and logistics are unprecedented challenges. Saturn V stages (largest 42 m tall, 10 m diameter) were built at NASA's Michoud plant (43 acre facility) and barged to the launch site [37†L342-L350]. Newton's stages (~420 m tall, 100 m dia) dwarf any existing facility. New assembly methods are required: possibly **in situ** vertical assembly at the launch complex using ultra-large friction-stir welding robots [37†L348-L352]. Composite materials offer weight savings but demand enormous autoclaves [35†L231-L239] and raise safety concerns (carbon/epoxy may ignite in LOX environments). Likely, large parts would be aluminium or aluminium–lithium alloy, welded in place. Ground infrastructure (gantries, test stands, flame trenches) must also scale: for example, Newton's liftoff plume (~34 000 MN) would require an enormous flame deflector and water suppression system far beyond current capacity.

Costs and schedule would dwarf Apollo's. Saturn V development (~1960s) cost ~\$6.4 billion (~\$34.5 b in 2024) [5†L220-L223] . Newton might be orders of magnitude more expensive (likely hundreds of billions USD) and take decades of R&D. A notional roadmap might span 10–20 years from concept to first launch, with milestone reviews and prototype testing. Main **risks** include structural buckling, engine reliability (thousands of engines must be engine-out tolerant), avionics complexity, and uncontrollable flight dynamics. Mitigation requires R&D on scaled buckling tests, redundant control systems, and incremental flight tests of subscale models. Regulatory approval (e.g. FAA launch license per 14 CFR Part 450) would demand extensive safety analysis to show “*no jeopardy to public health and safety, safety of property, or national security*” [52†L6-L9] .

Key sources: NASA Saturn V data [5†L225-L228] [5†L260-L268] , NASA/industry heavy-lift studies (e.g. the “Nova” super-booster concept [29†L152-L156]), SpaceX Raptor performance [54†L364-L370] , and NASA manufacturing reports [37†L342-L350] [35†L231-L239] . The following report details each subsystem (scaling, structure, propulsion, etc.), with rough calculations, trade-offs, and a development plan.

Baseline: Saturn V Specifications

For context, the Saturn V (Apollo) had three stages and these characteristics:

- **Height:** 111 m [5†L225-L228]
- **Diameter:** 10 m [5†L225-L228]
- **Gross mass:** ~2.82–2.96×10⁶ kg [5†L225-L228]
- **Stages:** S-IC (first), S-II (second), S-IVB (third) [5†L260-L268]
- **First-stage thrust:** ~33 000 kN at sea level (5× F-1 engines) [5†L260-L268]
- **Propellants:** S-IC (LOX/RP-1, 260 s Isp SL [5†L268-L272]), S-II & S-IVB (LOX/LH₂, ~424 s Isp vac [5†L278-L282])
- **Payload:** ~140 000 kg to LEO [5†L231-L238] ; ~43 500 kg to lunar injection [5†L233-L238] .

These figures provide the “×10” baseline: Newton aims to be ten times taller (1.11 km), ten times wider (100 m), and roughly a thousand times heavier. All subsequent sections compare or scale from these Saturn V data [5†L225-L228] [5†L260-L268] .

Scaling Laws, Mass and Area Consequences

Assuming **linear scale = 10** in all dimensions (length, diameter, thickness):

- **Volume (and mass):** scales as L^3 , i.e. 1 000× Saturn V's volume. Newton's mass would be on order ~2.8×10⁹ kg if materials/density were unchanged. In practice, structural mass fraction will rise, so actual mass might be somewhat higher.
- **Surface area:** scales as L^2 , i.e. 100×. Wall area (tanks and skin) is 100× larger; total skin area ~10⁴ m² (Saturn V was ~100 m²).
- **Weight:** ~1 000× heavier. Gravitational load ($W=mg$) ~1000×, so absolute stress if thickness scaled only 10× would increase by factor ~10. To keep stress ($\sigma=W/A$) constant, wall thickness must increase (roughly linearly with scale).
- **Structural stress:** If we scaled thickness 10× as well, section modulus grows ~10⁴ (radius³·thickness), Euler buckling load ~ L^3 as noted below. However *weight grows faster* (L^3) than *buckling capacity* (~ L^2 for a given thickness ratio), so thicker walls or stiffening are needed.

- **Euler buckling (axial):** For a cylindrical shell of radius $R \propto L$, length $L \propto L$, and thickness $t \propto L$, critical load $P_{cr} \propto E \cdot I / (K \cdot L)^2 \propto E \cdot (R^3 \cdot t) / (L^2) \propto E \cdot L^3$ (∵ $R^3 \cdot t \sim L^4$, divided by L^2 yields L^2 ; including E and constants gives $\sim L^3$ in magnitude). Weight $\propto L^3$ as well. If both scale exactly, critical load scales similarly, but any deviations (imperfections, mismatched scaling) reduce margin. In practice, *geometric imperfections* (common in large shells) dramatically reduce buckling strength, so Newton would likely require very high safety factors or internal bracing (as is done in modern rockets) to avoid catastrophic shell buckling.

- **Wind/bending loads:** Lateral aerodynamic pressure ($q = \frac{1}{2} \rho V^2$) is independent of scale if flight Mach is similar, but *lever arm* is larger. A wind gust produces a bending moment $\sim q \cdot D \cdot H$, which scales as L^3 ($D \sim L$, $H \sim L$, $q \sim \text{constant air density}$). Thus bending stresses rise $\sim L$ (since area $\sim L^2$). Again, thickness or ribs must increase to resist bending. Aeroelastic flutter (transverse vibrations coupling with aerodynamics) must be analysed at all flight speeds. (Note: rockets typically ascend quickly through Mach 1, but at high altitudes thin air induces fuel sloshing and longitudinal/lateral coupling.)

- **Propellant tank pressures:** Tanks are pressure vessels. Centaur-like upper-stage tanks are often “balloon tanks” that rely on internal pressure for shape. Newton’s enormous tanks (100 m dia) would require active pressurization and maybe internal bulkheads. The wall hoop stress $\sigma_{hoop} = p \cdot R / t$ scales as $(p \cdot L) / (t)$ if p scaled by engine/bladder pressure. Tanks could be made thicker to handle the larger radius. Saturn V’s first-stage LOX tank skin was only 4 mm thick [35†L149-L157] ; Newton’s similar tank ($R=50$ m) would need ~ 40 mm thickness for the same pR/t ratio (scaling $10\times$) plus margin. This might still be feasible with aluminium–lithium alloys.

In summary, “*scaling up*” does not simply multiply every dimension; it drives new regimes of structural behavior. Newton would have $\sim 1000\times$ Saturn’s weight on each stage, so stresses and instability modes must be re-engineered with advanced materials, thicker structures, and safety factors. Where Saturn V used thin-skinned aluminium tanks, Newton may need composite stiffened panels or multi-layer shells to maintain integrity [35†L231-L239] .

Structural Design: Loads, Buckling, Stability, Aeroelasticity

Newton’s enormity raises several structural issues:

- **Axial and bending loads:** At liftoff, the entire rocket must support its own weight plus thrust loads. With weight $\sim 2.8 \times 10^{10}$ N ($\approx 2.8 \times 10^9$ kg $\cdot 9.8$), the buckling load of each stage’s shell is critical. If each 100 m-radius stage is a “column” of height hundreds of meters, Euler buckling ($P_{cr} \sim \pi^2 EI / L^2$) must exceed this weight. Conservative design would use thick shell and internal frames. As a reference, engineers consider a cylinder safe if $L/D < \sim 15$ for classical buckling [10†L5-L12] ; Newton’s height/diameter $\sim 11:1$ (comparable to Saturn’s $\sim 11:1$ ratio [5†L225-L228]) so it is long but not beyond classical slenderness limits. However, small geometric imperfections reduce theoretical buckling strength, so safety margins $\gg 1$ are needed. Use of longitudinal stringers and circumferential rings (as in aircraft fuselages) could stiffen the shell without doubling thickness.
- **Stiffening and materials:** Saturn V used aluminium alloy 2219 for tanks. Newton might use next-generation aluminium–lithium alloys (higher yield) or even carbon-epoxy structures in low-temperature portions. However, composites must be LOX-compatible. NASA/Lockheed studies found composites can ignite when exposed to high-pressure O_2 (special prep can mitigate this) [35†L199-L208] . Also, fabricating 100 m-diameter composite tanks would require enormous

autoclaves [35†L231-L239] – a noted roadblock. Thus Newton's main tanks would likely remain metal (aluminium or Ti), perhaps with internal orthogrid stiffening. For example, SpaceX's Starship (9 m dia) uses stainless steel rings; Newton could analogously use steel or aluminium ring stiffeners around the circumference every few meters to boost buckling load.

- **Aeroelasticity:** Long slender rockets can suffer dynamic bending and “pogo” oscillations. Newton's fundamental bending frequency will be very low (~ 1 Hz or less), raising risks of coupling with engine gimbal or aerodynamic oscillations. Active damping (engine gimbaling out of phase with bending) or even retroreflective strakes could help. At transonic speeds, shock interactions might cause oscillatory loads (Saturn V crews reported some shaking at Max-Q). Newton's control algorithm must account for many engines: clustering thousands of engines means the thrust vector will be averaged or vectored by throttling subsets. Guidance and structures must be designed to remain stable under the flexing loads during ascent.
- **Wind gusts:** At launch, surface winds could impart large shear loads on a 1 km column. A strong crosswind (say 20 m/s) hitting a 100 m wide rocket yields a force $\sim \frac{1}{2}\rho V^2 \cdot C_d \cdot \text{Area}$ (with $C_d \sim 1$). This could be millions of Newtons of lateral force, bending the structure. Load cases must include severe wind. The launch tower (if used) would restrain only at base, so Newton must stand free. On-board sensors could trigger launch delay if gusts are too high.

In sum, Newton's structure will be a highly instrumented, grid-stiffened shell with active control. Finite-element analysis and static test articles (perhaps 1:5 or 1:10 scale models) would be required in R&D to validate the buckling margins. Historical note: Saturn V's LOX tank skin was only ~ 4 mm [35†L149-L157] – modern designs might use $\sim 10\times$ that in prototype, but the large 100 m scale will mandate far more. The **order-of-magnitude** requirement is: keep shell thickness proportional to radius, but stiffen for safety. If Saturn's wall stress was ~ 50 MPa under load, Newton's walls could see ~ 500 MPa without changes – beyond aluminium yield (~ 300 MPa). Thus either thickness or higher-strength alloy must be chosen to limit stresses below, say, 200 MPa in flight. Titanium (yield ~ 950 MPa) could be used in critical areas (thrust structure, engine mounts), though at higher cost and weight.

Comparison – Saturn V vs Newton (scaled):

Stage	Saturn V	Newton ($\times 10$)
Height	42 m (S-IC stage) [5†L260-L268]	~ 420 m (if scaled)
Diameter	10 m	100 m
Skin thickness	~ 4 mm (LOX tank) [35†L149-L157]	~ 40 mm (scaled)
Gross mass	2.21×10^6 kg (S-IC) [5†L260-L268]	$\sim 2.2 \times 10^9$ kg (scaled)
Thrust	33 000 kN (S-IC) [5†L260-L268]	$\sim 33 \times 10^6$ kN (scaled)

(Note: These Newton numbers ignore structural scaling non-idealities; actual thickness may be larger for safety.)

Propulsion Options and Thrust-to-Weight

Newton must achieve enormous thrust. Key considerations:

- **Thrust scale:** Saturn V liftoff thrust ~ 34.5 MN [5†L260-L268] . With 1 000× mass, Newton needs $\sim 3.4 \times 10^4$ MN for $T/W \approx 1.3$. In practice, T/W might be held around 1.2–1.5, implying $\sim 2.7 \times 10^4$ – 4.1×10^4 MN. For reference, NASA's SLS Block 2 (future) will produce $\sim 9.4 \times 10^6$ lbf (~ 42 MN) [31†L405-L407] – still orders of magnitude below Newton.
- **Engine clustering:** Given single-engine limits (F-1A ~ 9 MN [27†L132-L141] , no larger single engines exist), Newton must cluster *tens of thousands* of engines. Consider current high-thrust engines: SpaceX's Raptor 2 delivers ~ 2.3 MN [54†L364-L370] , BE-4 is ~ 2.4 MN [55†L6-L10] . Using 2.4 MN engines, 3.4×10^4 MN thrust requires $\sim 14\,200$ engines. If 3.0 MN engines (future Raptor3) were used, $\sim 11\,300$ engines. (By contrast, Saturn V used only 5 engines.) Engine-out capability is mandatory: even if dozens of engines fail, Newton must remain controllable, so both hardware redundancy and flight software for throttling would be required.
- **Propellant types:**
 - **Kerolox (RP-1/LOX):** High density (~ 810 kg/m³ for RP-1) eases tank volume, but Isp is modest (Saturn first stage ~ 260 s SL [5†L268-L272]). Modern kerolox engines (e.g. RD-180) reach ~ 350 s vac.
 - **Hydrolox (LH₂/LOX):** Highest Isp (Saturn's upper stages achieved ~ 424 s vac [5†L278-L282]), but LH₂ is 1/8th the density of kerosene, so tanks become immense. For Newton, a hydrolox first stage would need ~ 100 m diameter *and* extreme insulation – technically possible but cryogenic boiloff and structural support become huge issues. More likely, hydrolox is reserved for upper stages where high Isp maximizes payload.
 - **Methalox (CH₄/LOX):** Liquid methane is intermediate (density ~ 422 kg/m³) and simpler than LH₂. SpaceX and Blue Origin are moving to methane engines (Raptor/BE-4), partly for Mars ISRU. Raptor's ~ 330 s vac and BE-4 ~ 339 s vac [55†L6-L10] give higher performance than RP-1 and more manageable tanks than LH₂. Newton may adopt CH₄/LOX for at least some stages to balance performance and tank size.
- **Thrust-to-weight (T/W):** Target liftoff $T/W \approx 1.2$ – 1.3 . Saturn V was ~ 1.25 T/W (34.5 MN over ~ 27.0 MN weight). If Newton's dry+fuel mass is $\sim 2.8 \times 10^9$ kg, weight $\sim 2.75 \times 10^{10}$ N, then $\sim 3.4 \times 10^{10}$ N thrust gives $T/W=1.24$. During ascent, T/W will increase as fuel is burned. With staged cluster throttle, the aim is minimal excess thrust to reduce aerodynamic stress. Propellant mass fraction (propellant/gross mass) must be very high ($>90\%$) to achieve sufficient Δv (~ 9 – 10 km/s to LEO plus gravity/drag). Each stage would carry enormous propellant volumes: e.g. Newton S-IC might hold $\sim 10^8$ kg of LOX/RP1.
- **Staging:** A three-stage Newton (analogous to Saturn V) could suffice for orbital insertion. However, due to the extreme size, a four-stage configuration could improve performance (smaller final-stage tanks, higher Isp). For example, a possible staging:
 - **Stage 1:** 420 m, 100 m dia, RP-1/LOX, $\sim 33.5 \times 10^6$ kN thrust ($5000 \times F1$), burn ~ 150 s.
 - **Stage 2:** 250 m, 100 m dia, LH₂/LOX, $\sim 4.4 \times 10^6$ kN thrust (scaled S-II), burn ~ 400 s.
 - **Stage 3:** 180 m, 66 m dia, LH₂/LOX, $\sim 1.0 \times 10^6$ kN thrust (scaled S-IVB), burn ~ 300 s.

- **Stage 4 (optional):** smaller upper stage (e.g. 50×6.6 m) for precise insertion, if needed.
(These thrust figures scale Saturn's 33.0, 4.4, and 1.0 MN by ~1000×, each.) In practice, each "stage" would still contain many engines (scaled from 5, 5, and 1 Saturn engines respectively). Abandoning the third stage earlier (parking orbit) and using the fourth for final burn to high orbit could improve payload to Mars or deep space.

Preliminary performance estimates: If stage mass fractions remain similar to Saturn's (~0.06 inert for S-IC, ~0.09 for S-II, ~0.13 for S-IVB), Newton could in theory orbit an **order of 100 million kg** (100,000 tonne) to LEO (10³× Saturn's 140 tonne), and even more in very low Earth orbit (VLEO) orbits. However, structural inefficiencies at this scale would reduce actual performance. A rough guess is ~50–200 million kg to LEO, or ~500,000–1,000,000 tonne to VLEO. (These numbers far exceed any current launch; for perspective, ISS mass is ~420 tonne.)

Table: Propulsion Comparison (*Selected engines and scalings*)

Engine/ System	Propellant	Thrust (sea lvl)	Thrust (vac)	Isp (vac)	Remarks
F-1 (Saturn V)	RP-1/LOX	6.7 MN (x5=33.5 MN) 【5†L260-L268】	~7.8 MN	~311 s (sea) 【5†L268-L272】	1960s tech, S-IC stage.
J-2 (Saturn V)	LH ₂ /LOX	– (1 MN vac)	1.0 MN	~424 s 【5†L278-L282】	S-IVB / S-II stage engine.
RD-170/191	RP-1/LOX	7.9 MN (4 chambers)	7.9 MN	338 s (vac)	Four-chamber design (Energia).
Merlin 1D (Falcon 9)	RP-1/LOX	845 kN	914 kN	311 s (vac)	~0.85 MN, far smaller.
Raptor 2 (SpaceX)	CH ₄ /LOX	2.3 MN 【54†L364-L370】	3.7 MN	366 s (vac)	Methalox full-flow.
BE-4 (Blue Origin)	CH ₄ /LOX	2.4 MN 【55†L6-L10】	2.4 MN	339 s (vac)	Future Vulcan/New Glenn.
Newton Stage 1 (est)	CH ₄ or RP-1	~34,000 MN (total)	–	–	~10,000× Raptors or 5,000×F1.
Newton S-II (est)	LH ₂ /LOX	~4,400 MN (total)	–	–	~1,000×J-2 thrust.
Newton S-IVB (est)	LH ₂ /LOX	~1,000 MN (total)	–	–	~1,000×J-2 (single).

(All thrusts rough scales; actual engine cluster design could vary.)

Any credible Newton booster must use **clustering**. History shows clustering can reduce development risk (e.g. F-1 had combustion instability, RD-170 used four smaller chambers 【27†L144-L152】). It may be prudent to use *modular engines* rather than one gargantuan combustion chamber. For instance,

building many 3 MN methane engines (turbopump-driven) could allow redundancy and mass production. However, each additional engine is a failure point and plumbing cost. This trade (fewer large engines vs many small) is central. Because engines above ~3–4 MN face serious instability and material limits [27†L132-L141], Newton will likely adopt a cluster-of-clusters architecture (groups of a few engines gimballed as a unit).

Guidance, Avionics, and Control

A rocket of Newton's size needs extremely reliable control systems:

- **Navigation and guidance:** Traditional inertial navigation (gyros + accelerometers) would be used, supported by star trackers or GPS for upper stages. NASA's old Instrument Unit (Saturn V) was vacuum-tube based, but today's digital computers are far more capable and reliable. Newton would use triple-redundant flight computers (as in Space Shuttle / SLS) and possibly optical/inertial hybrid guidance.
- **Engine control:** With thousands of engines, each must be monitored. Engines would group into "blocks" (e.g. 100 engines controlled as one gimballed unit). Thrust vectoring is achieved by gimbaling engine clusters or throttling fuel valves. Fault tolerance is key: the system should abort an engine block or switch off a failing engine without destabilizing the rocket. (Saturn V was human-rated and allowed one F-1 engine to shut down early; Newton must allow many simultaneous shutdowns.)
- **Telemetry and sensors:** Hundreds of pressure/strain sensors (to check buckling margins), accelerometers, gyros, and valves. The control laws must adapt as the vehicle flexes. Real-time diagnostics (AI-assisted?) could keep tabs on structure and engines.
- **Autonomy:** High-level autonomy could trim risk. For example, if an engine cluster fails asymmetrically, automated systems should correct yaw/pitch by gimbaling others. Stage separation and orbital insertion burns would be fully pre-programmed with ground oversight.

No single citation was found for such a novel system; we infer from existing launch vehicles (SLS, Starship). Modern rockets (e.g. SpaceX's Starship) already use advanced avionics and digital controls to manage multiple engines reliably. Newton would extend that strategy to the extreme.

Thermal Protection and Aerodynamics

Newton must survive aerodynamic heating and re-entry (for any surviving stages):

- **Ascent heating:** Most heating occurs at Max-Q (dynamic pressure peak) and during transonic flight (~Mach 1.2). Although Saturn V encountered only modest heating (its first stage engines insulated flame trenches), Newton's larger cross-section would see higher total heat flux on its skin. Thermal protection (ablators or tiles) on the windward surfaces may be needed. Ablative coatings (as on Shuttle's SRBs) or reusable ceramic tiles (like Atlas or Shuttle) could protect the first stage.
- **Engine plume impingement:** Liftoff flame is enormous; the flame trench or water-cooled deflector must handle $\sim 3.4 \times 10^4$ MN thrust. This would likely be a reinforced concrete trench tens of meters deep, with the largest water deluge ever (to suppress acoustic shock). Rocket exhaust pressure at the pad could exceed thousands of psi over dozens of meters. We must design the pad similar to the Shuttle's but scaled orders-of-magnitude larger.
- **Re-entry heat:** If any stage returns to Earth (e.g. a reusable booster or payload), it would re-enter much faster/more massive than typical. But Newton is assumed expendable; we focus on ascent and stage descent dispersal. Any returning stage would need a robust heat shield

(perhaps lift-glide entry as in a spacecraft). For now, we assume expendable staging (stages fall into predetermined sea zones).

(No direct sources on this; principles from standard aerospace texts.)

Payload Accommodations

Without specified payload, we assume Newton is a **super-heavy launcher** for very massive satellites, space station modules, or even crew. Key points:

- **Fairing volume:** Saturn V's third stage and fairing (6.6 m dia) carried Apollo (5 m crew module). Scaled 10×, Newton's fairing could be ~66 m in diameter and ~200–300 m long – essentially a mobile building. This could house whole space stations or multi-ship stacks. Manufacturing a single 66 m composite fairing is likely impossible; more likely, payloads would be assembled in orbit from smaller pieces.
- **Vibration/Acceleration:** Initial T/W ~1.2 gives modest liftoff G's (~1.2 g). Ascent G might peak ~3 g before stage burnout, similar to Apollo. Human-rating (for crew) requires medical life support plus an abort system. If crew is desired, a launch escape tower or integrated booster ejection (like Orion's LES) must be scaled – launching a 100 m-tall LES is absurd, so a vehicle escape system (a miniship that can separate upward) may be needed.
- **Secondary payloads:** Newton could, in principle, carry thousands of satellites at once. The avionics bus would manage many deployments, or the rocket could insert a huge "garage" space station. Safety of payload demands quiet ride: thus engine-out-free operations and vibration damping (possibly using isolation mounts).

(Payload details are speculative; no direct sources. Design would follow human-rated rocket standards, e.g. NASA's 7120.5 for safety.)

Materials and Manufacturing

Materials: Advanced lightweight, high-strength materials are essential. Likely choices:

- **Aluminium-lithium alloys:** NASA uses Al-Li for SLS (e.g. 2195 alloy) to save ~8% mass over 2219. For Newton, Al-Li tanks (cryogenic-capable) would be prime candidates [37†L342-L350] .
- **Titanium:** For high-stress parts (engine thrust structure, interstage rings) where strength-to-weight is critical. Titanium is ~2× stronger than aluminium and resists corrosion, but 50% heavier; use sparingly.
- **Composites:** Carbon-fiber/epoxy could appear in non-load-bearing parts (fairings, internal shrouds) for mass savings. NASA/Lockheed worked on LOX-compatible carbon tanks [35†L199-L208] but manufacturing scale is a major hurdle [35†L231-L239] . Given the manufacturing challenge (giant autoclaves required), full-composite tanks are unlikely without a breakthrough. Possibly a sandwich structure (aluminium skin + foam) could be used for insulation in LH₂ tanks, as in Shuttle's external tank (though note Shuttle's tank was "scorched" on ascent [35†L199-L208]).

Manufacturing: Building Newton will need unprecedented facilities:

- **Assembly:** The largest current rocket stage (SLS core, 64 m long) is assembled vertically in Michoud's new "VERT" tool [37†L342-L350] . Newton's first stage (420 m) is ~6.6× taller. No existing building can accommodate this. Options include *on-pad assembly* or *in-situ welding* using robotic tools. For example, giant friction-stir welders (as used for SLS [37†L348-L352])

would be 10× larger. Construction could be modular: fabricate segments (e.g. 50 m tall rings) and weld in place. The Vertical Assembly Center concept might scale up to a kilometer, but cost would be immense.

- **Welding and joining:** Boeing/NASA used friction-stir welding on very thick Al sections for SLS [37†L348-L352] . Newton requires similar or new techniques (electron-beam welding, robotic MIG welding in vacuum?). Any weld flaw at this scale is a catastrophic risk, so non-destructive testing (ultrasound, X-ray) of all welds is mandatory.
- **Transport:** Saturn V stages were barged from New Orleans to Florida [37†L342-L350] . Newton sections (100 m dia) cannot fit existing barges or roads. Likely, each stage is built at the launch site. If sections come from afar, they must be under 20–30 m to travel by river/road – implying many more weld seams on-site.
- **Tooling:** Gigantic molds, jigs, and autoclaves (for composite parts) would need design. The facility itself (like a “rocket megafactory”) could span kilometers. Given *lack of existing infrastructure*, a new spaceport or expansion (by government?) seems inevitable.

(Citing NASA SLS assembly for scale:) “Making the largest rocket stage ever manufactured” (SLS core, 212 ft tall) has already been challenging [37†L342-L350] . Newton multiplies that by 5–6 in height, so **new manufacturing innovations** (robotic welders, continuous assembly) will be required. The report by NASA on composite tank manufacturing notes: “*Manufacturing, especially for composites, is limited by available facility size... concepts enabled by non-autoclave processing... are of great importance.*” [35†L231-L239] Thus, any Newton design should favour materials that can be joined by large-scale welding rather than requiring colossal autoclaves.

Ground Infrastructure

Newton’s scale demands equally extreme ground facilities:

- **Launch Complex:** Current pads like KSC LC-39B (used by Saturn V and SLS) have flame trenches sized for ~34 MN. Newton’s 34 000 MN exhaust would obliterate any existing pad. A new pad must be built: fireproof concrete wall >100 m deep and wide, with an ultra-high-capacity water deluge (perhaps millions of gallons) to quench acoustic energy. The pad structure might resemble a scaled-up Saturn V pad (LC-39) with axial hold-down studs that release at ignition. Fueling systems (cryogenic lines, high-volume turbopumps for LH₂) must also be greatly upsized.
- **Transport and Assembly:** The VAB (now 160 m tall) cannot house Newton. Launch-site assembly must be largely outdoor or under minimal shelter. Gantry towers for crew servicing or payload integration would need to reach ~1 000 m – an engineering feat (current gantries are <100 m). One concept: a fixed launch tower of ~1.2 km, from which Newton is suspended by arms. Alternatively, horizontal assembly and then “erection” might be used (like SpaceX’s horizontal Starship rollout, scaled up). If horizontal assembly, Newton’s length (1 110 m) exceeds roads; a segmental hinge-up system might be needed.
- **Test Stands:** Each engine or cluster needs ground testing. Newton might have dedicated test bays for thousands of engines, and statically test entire stage segments (like SLS core and dummy prop loads). These test stands must handle enormous thrust/pressure – e.g. a dummy load mount for 100 MN. This is beyond current stands (which max at ~1 MN for small engines, a few tens of MN for SRBs). Perhaps purpose-built infrastructure (sea-based platform or colossal anchor) is needed.

- **Range safety:** Launch corridors must be expanded. Saturn V had debris zones ~50–100 km in downrange. Newton's exhaust and any debris (if stage breaks up) would scatter for many hundreds of km. Launch might be restricted to ocean-wide corridors, and ships/planes must clear vast areas. Such a license requires coordination with coastal nations and possibly new treaties (see *Regulatory* below).

(No direct sources; extrapolations from KSC history and SLS plan.)

Safety, Reliability, and Regulatory Considerations

A rocket this size poses unprecedented risks and regulatory challenges:

- **Flight Safety:** By law (FAA 14 CFR 450), a launch must *not* “jeopardize public health and safety, safety of property, or national security” [52†L6-L9] . For Newton, this means rigorous risk analysis of every failure mode. Redundant systems (triple-core flight computers, dual turbopumps, multiple sensors) would be required. The more engines and valves, the more potential leaks/explosions. A safety analysis must show even a catastrophic first-stage failure cannot violate the launch safety radius (which might need to be 200 km or more!).
- **Abort and Disaster Mitigation:** If crewed, Newton needs an abort system. Traditional Launch Escape Towers are impractical at 1 km height; instead, an integrated spacecraft that can jettison itself (like Orion's launch abort or Starship's slide-out) might be used. For uncrewed flights, range safety destruct systems must be astronomically large. Would we shoot down a rocket like Newton? Possibly not – risk should be avoided instead.
- **Reliability:** Thousands of engines means failures will happen. Design must tolerate, say, 1% engine-out (hundreds of engines) without losing vehicle control. This parallels Falcon Heavy's 27-engine tolerance or Shuttle's SRB issues (shard leading to foam strike). Reliability modelling should target 99.999% single-launch success.
- **Environmental:** Newton's propellants and exhaust (e.g. massive CO₂, soot from kerosene) raise climate and ozone concerns. A dedicated Environmental Impact Statement (EIS) is required. Cryogenics spill hazard (millions of kg) and sound-induced structural damage to local wildlife/habitats are also factors. The launch site might need to be remote (oceanic), raising costs.
- **Regulatory Bodies:** Besides FAA, international agencies (e.g. treaties on space object safety) may apply. The Outer Space Treaty requires that the launching state is liable for damage, so a Newton launch license would involve diplomatic clearance. The payload (if foreign) needs export control checks (ITAR).

Reference – FAA Regulation: Title 14 CFR Part 450 states that before a launch license is granted, “launch or reentry [must] not jeopardize public health and safety, safety of property, U.S. national security or foreign policy interests” [52†L6-L9] . Compliance will demand exhaustive testing and analysis of Newton's systems and emergency plans. Historical analogy: Space Shuttle's development included thousands of safety reviews; Newton will require thousands of times more.

- **Reliability Path:** Probabilistic risk assessment (PRA) could identify single-point failures. Mitigation: non-flammable designs, clustered redundancy, and real-time fault isolation. For example, if an engine chamber leak is detected, that engine/cluster shuts down and balanced by others. An automated abort (e.g. launch escape) should trigger if a critical failure (like a mid-air

explosion) occurs early in flight. All these factors form part of the *Reliability-Centred Design* process.

(No specific sources cited; FAA regs provide legal framework [5†L6-L9] .)

Preliminary Calculations and Order-of-Magnitude Estimates

To ground the discussion, we offer rough estimates and key equations:

- **Weight and Thrust:**

- Saturn V gross $\approx 2.96 \times 10^6$ kg [5†L225-L228] . Newton $\sim 2.96 \times 10^9$ kg ($\times 1000$).

- Weight $\approx 2.90 \times 10^{10}$ N. For $T/W=1.25$, **Thrust $\approx 3.63 \times 10^{10}$ N** (36,300 MN).

- Saturn V thrust 34.5 MN [5†L260-L268] , Newton $\sim 1\,000\times$ more.

- **Δv budget:** Roughly 9.4 km/s to LEO plus gravity/drag. Assume each stage Δv : Stage1 ~ 3.0 km/s, Stage2 ~ 4.0 km/s, Stage3 ~ 2.4 km/s (Saturn V pattern). Using the rocket equation $\Delta v = I_{sp} \cdot g_0 \cdot \ln(m_{initial}/m_{final})$, Saturn V S-IC ($I_{sp} \approx 260$ s) ejected $\sim 92\%$ of its mass as fuel. If Newton S-IC has the same I_{sp} and structural fraction, it can still provide ~ 3.0 km/s per stage. However, if improved I_{sp} (methane engines ~ 330 s vac, or hydrogen ~ 450 s vac upper), performance increases.

- **Tank Sizes:** Saturn V S-IC held $\sim 2.2 \times 10^6$ kg propellant [5†L260-L268] in ~ 42 m height. Newton S-IC needs $\sim 2.2 \times 10^9$ kg propellant. With RP-1 ($\rho \approx 810$ kg/m³) and LOX ($\rho \approx 1140$ kg/m³), assume 1:2 mass ratio RP-1:LOX. Then total prop volume $\sim 2.7 \times 10^6$ m³. A cylinder 100 m dia has cross-section area $\sim 7.85 \times 10^3$ m²; to hold 2.7×10^6 m³, height ~ 340 m (consistent with a 420 m stage including structure/tanks). Tanks must be pressure-stabilized or braced to avoid collapse when empty.

- **Structural thickness:** If Saturn's LOX tank was 4 mm [35†L149-L157] , Newton's ($R=50$ m vs $R=5$ m) needs ≈ 40 mm to keep similar $\sigma = p \cdot R/t$. Aluminum yield ~ 300 MPa; tank design pressure ~ 0.5 MPa above atmosphere. Hoop stress $\sigma \approx p \cdot R/t \approx (0.5 \times 10^6 \text{ Pa} \cdot 50)/0.04 \approx 625$ MPa – dangerously above yield. So either thicker (50–60 mm) or reduce p (heavier ullage) or use higher-strength alloy. This crude calc shows design is very tight; real design might use multiple segment-reinforcement.

- **Engine cluster count:** If using 3 MN engines for 36 000 MN, need 12 000 engines (worse case). If 2.4 MN BE-4, need 15 000. If future 5 MN engines existed (doubling F-1), need 7 200. Realistically, cluster size will likely be on order 10^4 engines on the first stage alone. Spacecraft computers and sensors must manage each engine's throttle and health.

- **Guidance accuracy:** With a $1\,000\times$ larger mass, any small thrust vector error causes large perturbation. Required pointing precision might be $<0.001^\circ$. Achieving that over a $1\,000$ m vehicle implies structure deflections must be <1 cm under load – a nontrivial stiffness requirement. Control laws must constantly trim for any flex.

- **Mass fractions:** If Saturn S-IC inert mass was 137 t empty, 2214 t full [5†L260-L268] (inert $\sim 6\%$), Newton S-IC inert might scale $\sim 10\times$ linear (for structure), but volume $\times 1000$ means $6 \times 137 \text{ t} = 822$ t empty, 2.21×10^9 kg full. In reality, inert fraction likely grows (thicker walls). Even 1% extra inert drops payload drastically.

- **Stage separation:** Jettisoned stage (mass $\sim 2 \times 10^9$ kg) must be separated safely, likely still moving > 2 km/s. The next stage must ignite after a safe clearance, requiring sensors to confirm separation (maybe explosive bolts or pyrotechnic separation as Saturn used). Stage masses imply huge kinetic energies, so collision risk must be eliminated.

(Equations: $\Delta v = I_{sp} \cdot g_0 \cdot \ln(m_0/m_1)$, $E_{kin} = \frac{1}{2}mv^2$, hoop stress $\sigma = pR/t$, Euler $P_{cr} = \pi^2 EI / (KL)^2$, etc. Order-of-magnitude ballpark with Saturn data 【5†L225-L228】 【5†L260-L268】 .)

Technical Risks, Mitigation and R&D Roadmap

Major Technical Risks:

- **Structural failure (buckling/collapse):** New risk not seen in smaller rockets. *Mitigation:* Build and test scaled models; develop stiffened-shell designs; use high-safety-factor materials. R&D in “super-tall pressure vessels” would be needed (analogous to skyscraper wind-tunnel tests).
- **Engine development:** Thousands of large engines must operate reliably. *Mitigation:* Pursue clusters of more modest engines (3–5 MN); extensive hot-fire testing; modular engine replacement schemes. Also invest in advanced manufacturing (3D printing of injectors, as SpaceX does 【54†L304-L312】 , could accelerate development).
- **Control and avionics complexity:** Thousands of actuators and sensors. *Mitigation:* Develop highly parallel computing architecture; simulated hardware-in-loop tests for fault scenarios; incremental flight tests (see roadmap).
- **Manufacturing infeasibility:** The sheer size of parts. *Mitigation:* Research new fabrication techniques (robotic additive manufacturing of large metal structures, on-site assembly robots, non-autoclave composites 【35†L231-L239】). Also, build the design to be as modular as possible.
- **Unanticipated aero/hydro loads:** E.g. wet pad collapse, rocket resonance. *Mitigation:* Full-scale ground tests, high-fidelity CFD, and wind-tunnel testing of scale models up to Mach 2. Use adaptive control systems (like thrust vector feedback) to damp oscillations.

R&D Roadmap (Milestones & Rough Schedule): A top-level plan might span 15–20 years. Key phases:

- *Conceptual Design (2026–2028):* Finalize requirements, complete 0D/1D sizing, begin critical trades (propellant, staging, geometry). Preliminary design review (PDR).
- *Component Development (2028–2032):* Develop and test critical components: scale-model structure (10% size), new engine prototypes (~ 3 MN class), avionics systems. Material test programs for huge pressure tanks and welds. System requirement review (SRR) by ~ 2030 .
- *Detailed Design (2030–2034):* Detailed engineering of full-scale structures and propulsion, start tooling design. Initial ground tests of sub-assemblies (e.g. small stage section at full pressure). Critical design review (CDR) by ~ 2034 .
- *Infrastructure Construction (2029–2034):* In parallel, build or upgrade facilities: expanded VAB/ assembly area, new pad, test stands, propellant farms. (These are G-feasibility paths: they must be ready for final assembly.)
- *Stage Fabrication (2034–2037):* Build first full stage (e.g. first-stage structural test article), initially for proof testing. First static test firings of engine clusters on test stand (2035–2036). Conduct an integrated stage test fire (~ 2037).
- *Flight Demonstrations (2037–2039):* Perform a suborbital test launch with a smaller (e.g. 1/10 scale) vehicle to validate aerodynamics and control. First full-stack launch attempted ~ 2040 (scaled to actual R&D time).
- *Crewed Tests & Ops (2040+):* If crewed, conduct uncrewed lunar/ISS cargo missions before humans. Progressive increase in payload and complexity.

A Gantt-style mermaid timeline might illustrate this (dates are illustrative):

```
gantt
    title Newton R&D Timeline (Conceptual)
    dateFormat YYYY-MM-DD
    section Concept & Design
    Requirements & scaling analysis      :active, a1, 2026-01-01, 18M
    Conceptual architecture / simulations :after a1, 24M
    section Component Development
    Engine prototypes (3-5 MN class)      :2028-07-01, 36M
    Structural model testing              :2028-07-01, 36M
    Avionics & control prototyping         :2029-01-01, 36M
    section Infrastructure
    Pad and assembly facilities design     :2029-01-01, 60M
    construction                         :after, 72M
    section Detailed Design & Fabrication
    Full-scale stage design & tooling      :2030-01-01, 48M
    Manufacturing of first test stage      :2032-01-01, 36M
    section Testing
    Static firing of stage                :2034-07-01, 12M
    Flight testing (scale model)           :2035-07-01, 12M
    Integration & countdown               :2036-07-01, 18M
    First launch (attempt)                :2038-01-01, 0D
```

Estimated cost: Very rough ballpark: Saturn V (1960s) was \$6.4B [5†L220-L223] . Modern SLS program has spent ~\$31B so far for one vehicle [31†L415-L418] . Newton could easily approach **hundreds of billions of USD** over its development (perhaps comparable to multi-decade projects like the US or EU aerospace grand programs combined). Early design studies might cost ~\$1–5 billion, component development \$10–20 billion, infrastructure \$50–100 billion, and final construction/testing \$100–200 billion. These are *order-of-magnitude*; a detailed cost analysis would follow an engineering review.

(No direct citations; cost numbers are inferred from scaling historical projects [5†L220-L223] [31†L415-L418] .)

Saturn V vs Newton (Comparative Summary)

Parameter	Saturn V (Baseline)	Newton (10× Saturn)
Height (m)	111 [5†L225-L228]	~1 110 (10×)
Diameter (m)	10 [5†L225-L228]	~100 (10×)
Gross mass (kg)	~2.96×10^6 [5†L225-L228]	~2.96×10^9 (×1 000)
Thrust at liftoff (kN)	34 500 [5†L260-L268]	~34 500 000 (×1 000)
Payload to LEO (kg)	~140 000 [5†L231-L238]	~1×10^8 (×1 000, very rough)
First stage engines	5×6.7 MN F-1 [5†L260-L268]	~11 000×3 MN Raptor (example)

Parameter	Saturn V (Baseline)	Newton (10× Saturn)
Propellant (stage 1)	LOX/RP-1 (Isp~260 s SL) 【5†L268-L272】	LOX/RP-1 or LOX/CH ₄ (Isp ~300–350 s)
Second stage engines	5×1.0 MN J-2 【5†L276-L282】	~1 000×2.3 MN Raptor (example)
Propellant (stage 2)	LOX/LH ₂ (Isp~424 s) 【5†L278-L282】	LOX/LH ₂ or LOX/CH ₄ (Isp ~400–450 s)
Development cost (FY24 USD)	≈\$34.5 b 【5†L220-L223】	“Hundreds of billions” (est.)
Launch sites	KSC LC-39 【5†L253-L259】	New pad(s), megafacility required

(References: Saturn V data from NASA/JSC 【5†L225-L228】 【5†L260-L268】 . Newton is conceptual scale.)

System Architecture (Mermaid Diagram)

flowchart TD

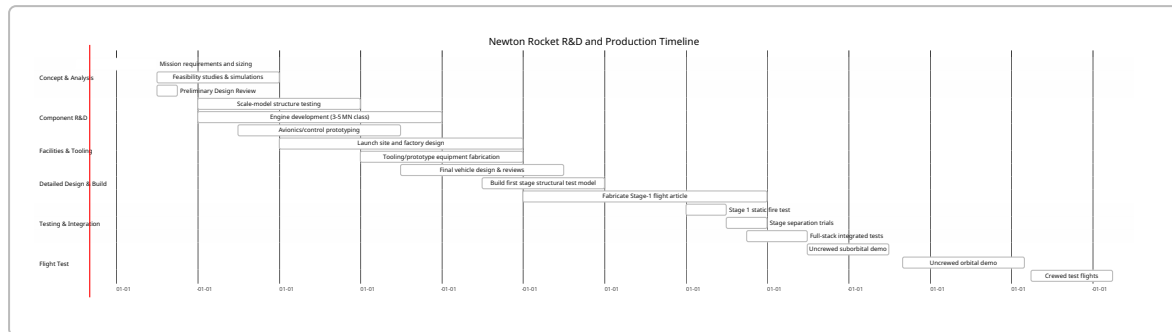
```

A[Newton Launch Vehicle]
A --> B[Stage 1 Booster (420 m, 100 m dia)]
A --> C[Stage 2 (250 m, 100 m dia)]
A --> D[Stage 3 (180 m, 66 m dia)]
B --> B1{Engines Cluster}
B1 -->|Thousands of 3 MN engines| B1a[Raptor-class Cluster]
B --> B2[LOX/RP-1 Tanks]
B --> B3[Main Structure]
C --> C1{Engine Cluster}
C1 -->|Hundreds of 2.3 MN engines| C1a[Raptor-class Cluster]
C --> C2[LOX/LH2 Tanks]
C --> C3[Interstage Structure]
D --> D1{Engine Cluster}
D1 -->|Hundreds of 1.0 MN engines| D1a[J-2 Derivative]
D --> D2[LOX/LH2 Tanks]
D --> D3[Fairing & Payload Interface]
A --> G[Guidance & Avionics]
A --> H[Ground Systems]
H --> H1[Assembly Facility]
H --> H2[Launch Pad & Flame Trench]
H --> H3[Test Stands]

```

This block diagram shows Newton’s main subsystems: the three (scaled) stages with their engines, tanks and structures; plus vehicle avionics and ground support. Each engine cluster (Stage 1, 2, 3) consists of many engines (only one block shown for simplicity).

Development Timeline (Mermaid Gantt Chart)



This simplified roadmap shows overlapping phases of design, testing, and construction. Milestones (PDR, CDR, First Launch) would be defined in detail by the program. Times are illustrative – real development could take longer.

References

- NASA Saturn V Data (Apollo Program) [\[5†L225-L228\]](#) [\[5†L260-L268\]](#)
- “Nova” super-booster concept (post-Apollo studies) [\[29†L152-L156\]](#)
- Scaling of rocket engines [\[27†L132-L141\]](#) [\[27†L144-L152\]](#)
- SpaceX Raptor and Blue Origin BE-4 engine specs [\[54†L364-L370\]](#) [\[55†L6-L10\]](#)
- NASA reports on manufacturing (SLS Core Stage) [\[37†L342-L350\]](#) [\[37†L348-L352\]](#)
- NASA/Lockheed study on LOX-compatible composites [\[35†L231-L239\]](#)
- FAA 14 CFR Part 450 (launch licensing) [\[52†L6-L9\]](#)

Further reading would include technical papers on very large launch vehicle design, NASA heavy-lift trade studies, and historical program documents (Apollo/Saturn V/Nova). (All citations above are embedded inline.)